Contents

1 Probability

Definition 1.0.1 ▸ Discrete Probability Model

Let Ω be a set , $|\Omega| < \infty$. $Q = \varphi(\Omega)$ the collection of all subsets of Ω .
$P : \varphi(E) \to [0,1]$ a function . If
1. $P(\Omega) = 1$.
2. $P(A \cup B) = P(A) + P(B)$ for all $A, B \in \varphi(\Omega)$, $A \cap B = \varnothing$.

Then we call (Ω, Q, P) a discrete probability model .

Proposition 1.0.2

Let (Ω, Q, P) be a discerte probability model ,
1. Let $A \in \varphi(\Omega)$, $\bar{A} := \{x \in \Omega, x \notin A\}$. THen
$$P\left(\bar{A}\right) = 1 - P(A)$$
2. $A, B \in \varphi(\Omega)$, then
$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Remark.
If $P(\{\cdot\}) = \frac{1}{|\Omega|}$, then we say the probability is ***uniform*** .

Example 1.0.3

Take $\Omega = \{a, b\}$, $Q = P(\Omega) = \{\varnothing, a, b, \Omega\}$. $P(a) = \theta, P(b) = 1 - \theta$. Then P can extends to a probability, and (Ω, Q, P) comes to a discerte probability model.

Definition 1.0.4 ▸ Random Variable

A ***Random Variable*** (R.V) is just a map $X : \Omega \to \mathbb{R}$.

Example 1.0.5 ▶ A Random Variable

$A \in \varphi(\Omega)$. Let $X = \chi_A$. Then X is a random variable. And

$$P(X = 1) = P(A), \quad P(X = 0) = 1 - P(A).$$

Definition 1.0.6 ▶ Expectation (Mean Value of a R.V)

Let X be a $R.V$. If

1. $|X(\Omega)| < \infty$ and $X \geq 0$. We define

$$E(X) := \sum_{x \in X(\Omega)} xP(X = x)$$

2. In general case, if $E(|X|) < \infty$, we define

$$E(X) := E(X^+) - E(X^-)$$

Proposition 1.0.7

Let $\varphi : X(\Omega) \to \mathbb{R}$. Assume that $E(|\varphi(X)|) < \infty$. Then ,

$$E(\varphi(x)) = \sum_{x \in X(\Omega)} \varphi(x) P(X = x)$$

Definition 1.0.8

Suppose that $E\left(X^2\right) < \infty$. Then define the ***variance*** of X

$$\mathrm{Var}(X) := E\left[(X - E[X])^2\right] = E\left[X^2\right] - (E[X])^2$$

Proof.

$$\begin{aligned}\mathrm{Var}\, X = E\left[(X - E[X])^2\right] &= E\left[X^2 - 2E(X)X + (E(X))^2\right] \\ &= E\left(X^2\right) + E(-2E(X)X) + E\left((E[X])^2\right)\end{aligned}$$

$\square$

Definition 1.0.9 ▸ Probability Generating Function

Let X be a discrete random variable with range in $\{0, 1, 2, \dots\}$. Its Probability Generating Function, $G_X(t)$, is defined as:

$$G_X(t) := E[t^X]$$

for all real numbers t for which the expectation exists. Specifically,

$$G_X(t) = \sum_{k=0}^{\infty} t^k P(X = k)$$

Proposition 1.0.10

Assume that $R > 1$, Then

1. $E(X) = G'_X(1)$
2. $\mathrm{Var}(X) = G''_X(1) + G'_X(1) - [G'_X(1)]^2$

Example 1.0.11 ▸ Poisson

Let $X \sim \text{Poisson}(\lambda)$,

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad \text{for } k = 0, 1, 2, \dots$$

$$\begin{aligned} G_X(t) = E[t^X] &= \sum_{k=0}^{\infty} t^k P(X = k) \\ &= \sum_{k=0}^{\infty} t^k \frac{\lambda^k e^{-\lambda}}{k!} \\ &= e^{-\lambda} \sum_{k=0}^{\infty} \frac{(t\lambda)^k}{k!} \quad \text{(将常数和 t 移入)} \end{aligned}$$

$$G_X(t) = e^{-\lambda} \cdot e^{t\lambda} = e^{t\lambda - \lambda} = e^{\lambda(t-1)}$$

$$G_X(t) = e^{\lambda(t-1)}$$

Index

Definitions

Examples

Propositions